

# Aniket Dev Roy

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## EDUCATION

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### Purdue University

*Master of Science in Geo-Data Sciences*

West Lafayette, IN

Aug. 2024 – Dec. 2025

### Purdue University

*GIS Certificate | Credential ID: 25I2-0NIR-AOYY*

West Lafayette, IN

2025

### Purdue University

*Bachelor of Science in Earth Sciences*

West Lafayette, IN

Aug. 2020 – May 2024

## EXPERIENCE

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### Graduate Assistant in GIS

*Purdue University Libraries*

Jan. 2025 – Present

West Lafayette, IN

- Performed multi-state food-access (food desert) analysis across census tracts, integrating Census TIGER/Line shapefiles, USDA Food Access Research Atlas data, and ACS demographics via the Census API in Python (GeoPandas). This is parameterized to replicate for any U.S. state.
- Built network-based accessibility workflows in QGIS (shortest-path routing, service-area generation on OpenStreetMap road networks), including CRS management, clipping, and geometry QA, to measure grocery-store access beyond straight-line distance.
- Cleaned and standardized raw multi-source geospatial datasets in ArcGIS Pro into analysis-ready teaching data, and used ArcGIS Pro as an independent check to validate map outputs produced in Python.
- Authored 6 open-source, Carpentries-based training modules covering the geospatial data lifecycle (collection, analysis, visualization, curation) for an NSF-funded Purdue cybertraining project, published publicly via GitHub Pages.
- Taught hands-on QGIS, remote sensing, and Python sessions at a 2-day Purdue–NSF GIS Cybertraining workshop for junior and senior GIS professionals from across Indiana.

### Undergraduate Seismic Researcher

*Purdue University College of Science*

May 2023 – Aug. 2024

West Lafayette, IN

- Benchmarked Principal Component Analysis (PCA) and a custom grid-search algorithm for estimating near-surface shear-wave velocities from earthquake particle motion, generating synthetic seismic datasets via Linux shell scripting across frequency, layer-thickness, and time-window parameter spaces.
- Diagnosed a systematic covariance bias in the PCA approach and validated a grid-search misfit-minimization alternative that delivered consistently low-error velocity estimates.
- Applied both methods to broadband data from 60 seismic stations in Northern California (16 quality-checked earthquakes), producing spatial maps of shear-wave velocity to flag areas of elevated earthquake-shaking hazard; manually picked P- and S-wave arrivals to build the ground-truth evaluation dataset.
- Co-authored an abstract presented at AGU Fall Meeting 2024; designed and presented a first-author poster at the Purdue Undergraduate Research Symposium. Tools: MATLAB, Python, Linux.

## PROJECTS

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### Global Tropical Cyclone Intensity Trends | *Python, SciPy, Cartopy*

Aug. 2025 – Dec. 2025

- Built an end-to-end Python pipeline analyzing 45 years (1980–2024) of global tropical cyclone records, automating download and preprocessing of NOAA IBTrACS best-track and HURSAT satellite NetCDF data into analysis-ready  $4^\circ \times 4^\circ$  gridded datasets (xarray, pandas, NumPy).
- Applied robust statistics like Theil–Sen estimators (SciPy) and bootstrap confidence intervals on regression slopes (statsmodels) with sensitivity tests across storm-strength thresholds and time periods; mapped results with Cartopy.
- Cross-validated trends between IBTrACS and the satellite-homogenized HURSAT dataset, finding intensification concentrated in major hurricanes ( $\geq 64$  kt), strongest in the North Atlantic; delivered a reusable, parameterized notebook and datasets via GitHub.

- Designed a multi-county spatial analysis in ArcGIS Pro investigating the correlation between ground-level ozone and COVID-19 infection rates across three climatically distinct U.S. counties (Marion IN, Los Angeles CA, Miami-Dade FL) using 2022 EPA air-quality and public-health data.
- Built the geoprocessing workflow end to end which included attribute selection, table joins, spatial joins, pairwise clip, buffer, and erase that integrated county demographics, ozone monitoring sites, and road networks; scripted a data-resaping step in Python (pandas).
- Produced choropleth maps, road-network-based medical-accessibility scores, and correlation scatterplots; published results as an interactive ArcGIS StoryMap for a general audience.
- Found regionally inconsistent ozone–COVID correlations (0.67 in Los Angeles, 0.36 in Marion, ~0 in Miami-Dade), attributing divergence to climate-driven seasonal ozone patterns and contextualizing against a prior single-region study.

## SKILLS

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**GIS Platforms:** ArcGIS Pro, QGIS, ArcGIS Online, ArcGIS StoryMaps, QField, Survey123, Experience Builder

**Programming:** Python (GeoPandas, pandas, NumPy, xarray, SciPy, Matplotlib, Cartopy), MATLAB, Linux/Bash, Git/GitHub

**Geospatial Data:** Census API/ACS, TIGER/Line, OpenStreetMap, USDA FARA, NetCDF, shapefiles, network analysis, CRS management

**Core:** Cartography & Thematic Mapping, Technical Documentation, Instruction & Training, Reproducible Workflows